

Round 3 Report — Autoregressive Forecasting

Overview

The objective of this task was to predict Geetha's typing performance for the next day (day 999) using historical data from the previous 999 days. The dataset consisted of four features: WPM, Accuracy, Raw WPM, and Consistency.

Data Preprocessing

Only the relevant four features were retained, and any unnecessary columns were removed.

A preliminary inspection of the data revealed short-term fluctuations, occasional spikes, and gradual trends in certain features. This indicated the presence of noise, suggesting that smoothing techniques could be beneficial.

Baseline Models

Two baseline approaches were implemented.

- The first was the Last Value Predictor, where the next value is assumed to be equal to the most recent observation.
- The second was the Moving Average method, where recent values are averaged to reduce short-term noise.

From the results, it was observed that the moving average performed better for Raw WPM and Consistency, while the last value predictor performed well for WPM. These baselines provided a useful reference point for further improvements.

ARIMA Model

An ARIMA (Autoregressive Integrated Moving Average) model with order (1,1,1) was applied independently to each feature.

This approach helped capture temporal dependencies more effectively, particularly for Accuracy, where trend-based patterns were observed.

However, performance varied across features, and ARIMA was not optimal for all of them, especially Raw WPM.

Model Optimization Strategy

Upon evaluating different approaches, it was observed that no single model performed best across all features.

Therefore, a feature-wise optimization strategy was adopted. Additionally, the penalty structure of the evaluation metric was carefully considered. Since Raw WPM has an exponential penalty and Consistency has a $4\times$ penalty, reducing errors in these features was prioritized.

Final Model (Ensemble Approach)

A hybrid model was constructed by selecting the best-performing method for each feature.

- WPM $\rightarrow$ Last value predictor
- Accuracy $\rightarrow$ ARIMA + recent value
- Raw WPM $\rightarrow$ Last value + median
- Consistency $\rightarrow$ Median

The use of the median instead of the mean helped in handling outliers and fluctuations effectively.

Results and Observations

After tuning and combining models, significant improvements were observed.

WPM prediction closely matched the actual value.
Consistency prediction achieved near-perfect accuracy.
Raw WPM error was significantly reduced compared to initial attempts.
Accuracy improved through the use of ARIMA.

Overall, the ensemble approach resulted in a balanced and optimized prediction across all features.

Conclusion

This task demonstrates that a combination of simple models and time-series techniques can yield strong results when applied thoughtfully.